Yashasvi Sorapalli

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EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Master of Science Robotic Systems Development — **May 2028**
Coursework Fall 2026: Introduction to Deep Learning; Robot Autonomy; Robot Mobility on Air, Land and Sea

RV College of Engineering, Affiliated with VTU — **Bangalore, India**
Bachelor of Engineering in Computer Science and Engineering | CGPA 9.06/10.00 — **July 2024**

SKILLS

Languages: Python, C++, Java, SQL **| Robotics & ML:** ROS 2, RViz, MuJoCo, PyTorch, NumPy, OpenCV, MATLAB **| Tools:** Linux, Git, AWS, SolidWorks

ACADEMIC PROJECTS

DARWIN - Autonomous Robotic System for Wet Lab Experiments — **Pittsburgh, PA**
Carnegie Mellon University, Capstone — **September 2026 - Present**

- Designing an autonomous laboratory system integrating 1 custom mobile base and a 6-DOF xArm6 robotic arm to execute language-driven experiments across 3+ workstations, including an Opentrons liquid handler, shaker, and spectrophotometer.
- Developing vision-based anomaly detection and recovery workflows for 2+ classes of labware, including misplaced and tilted well plates, using dynamic task replanning and human escalation when autonomous recovery fails.
- Architecting end-to-end experiment monitoring and reporting to log task execution, notify scientists of run outcomes, and route unsuccessful batches to a designated disposal location.

Beyond the Training Grid: Generalization and Uncertainty in Deep Energy Models for Delivery Drones — **Pittsburgh, PA**
Carnegie Mellon University, Introduction to Deep Learning, supported by Google — **September 2026 - Present**

- Porting a published TensorFlow TCN and LSTM power model to PyTorch to reproduce a 9.06% MAPE baseline on 195 real quadrotor flight logs.
- Building a physics-informed hybrid: an analytical power model predicts the trend and a small TCN learns the correction.
- Adding calibrated uncertainty targeting 90% prediction-interval coverage, and feeding it into a CVaR pre-flight score.

F1Tenth Autonomous Driving, Franka Manipulation & Vehicle Control — **Pittsburgh, PA**
Robot Autonomy; Robot Mobility, CMU course projects — **August - October 2026**

- Engineered five autonomous navigation algorithms using ROS 2, including time-to-collision safety braking, PID wall tracking, reactive gap selection, pure pursuit path following, and RRT trajectory planning; deploying four onto a physical 1/10-scale car.
- Tracked smoothed spline waypoints with pure pursuit on hardware using particle-filter localization, and ran RRT to plan around obstacles in real time in a hallway environment.
- Implemented forward kinematics for a 7-DOF Franka Panda arm in Python, and compared force and impedance control in MuJoCo for contact with fixed and moving sinusoidal walls.
- Designed an LQR lateral controller for a dynamic bicycle-model lane change in MATLAB, staying within about 1 cm of the desired path and meeting all specs.

PROFESSIONAL EXPERIENCE

Visa Inc. — **Bangalore, India**
Software Engineer — **September 2025 - August 2026**

- Built real-time Grafana dashboards combining Prometheus metrics and OpenSearch logs for 3 core services of VisaPay, giving the team a single view for on-call debugging.
- Supported the in-housing of VisaPay from an external vendor, resolving integration issues with vendor engineers and documenting the system for team handover.

BT Group — **Bangalore, India**
Software Engineering Professional — **July 2024 - September 2025**

- Developed AWS Lambda function automating PostgreSQL data cleanup, reducing stale data by 80% and improving reliability.
- Led stakeholder engagement efforts, successfully increasing user interest and adoption of an unused product by 7x.
- Ensured 100% uninterrupted operation of the internal analysis tool during a company wide restructuring.

BT Group — **Bangalore, India**
Graduate Engineering Trainee Intern — **January 2024 - July 2024**

- Engineered a React.js interface integrated with Node.js to work with production APIs, boosting cross-functional usability.
- Integrated Azure AD into frontend, enhancing system security protocols and user authentication control.

Indian Institute of Astrophysics — **Bangalore, India**
Software Intern — **October 2023 - January 2024**

- Automated the calculation of atmospheric distortion using high-frequency solar imaging data captured milliseconds apart.
- Constructed end-to-end pipeline using LabVIEW to extract brightness profile 8x1 matrices from dual-slit telescope images.

Multi-camera Multi-person Tracking using DeepSORT and MySQL — **Bangalore, India**
Undergraduate Research Intern, Inferigence Quotient — **September 2022 - August 2023**

- Engineered a multi-camera tracking architecture that scales DeepSORT across disjoint camera feeds using a centralized MySQL synchronization framework, reducing identity resets by 85% during prolonged occlusions.
- Optimized re-identification latency with indexed database schemas, enabling continuous tracking across broad visual fields.
- *Published paper: Multi-camera Multi-person Tracking using DeepSORT and MySQL, Vol. 38, No. 2 of IJEECS*